



Generating Functions and Percolation on Graphs

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Introduction to Generating Functions

An ordinary generating function is a formal power series $f(x) = \sum_{n=0}^{\infty} a_n x^n$ whose coefficients correspond to terms of a sequence $\{a_n\}$. Since generating functions are formal power series, they can be evaluated without considering issues of convergence and can be manipulated using properties such as addition and multiplication. One of the primary functions of generating functions is solving recurrence relations since they contain all the information of a sequence in a single series, whereas recurrence relations can only relate a few terms of the sequence.

Interesting Examples of Generating Functions

We can use generating functions to create a closed form expression for the **Fibonacci numbers**, which are defined by $F_0 = 0, F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. To get this expression, first we let $f(x) = \sum_{n=0}^{\infty} F_n x^n$. We multiply both sides of the recurrence relation by x^n and sum over n to obtain:

$$\sum_{n=0}^{\infty} F_{n+2} x^n = \sum_{n=0}^{\infty} F_{n+1} x^n + \sum_{n=0}^{\infty} F_n x^n$$

$$\frac{1}{x^2}(f(x) - x) = \frac{1}{x}f(x) + f(x)$$

Solving this equation gives us the closed form expression for the Fibonacci sequence:

$$f(x) = \frac{x}{1-x-x^2}.$$

The coefficient operator $[x^k]$ extracts the coefficient of x^k in a formal power series in x so $[x^k]A(x) = a_k$ when $A(x) = \sum_{n=0}^{\infty} a_n x^n$. Thus after computing the power series expansion of $f(x)$, we find F_n by extracting the coefficient on x^n .

As another application of generating functions, we will count the number of binary trees with $n+1$ leaves, denoting this number by C_n . From this definition, we obtain the following recurrence relation:

$$C_0 = 1; C_{n+1} = \sum_{k=0}^n C_k C_{n-k}$$

The sequence this creates is called the **Catalan Numbers**, and they appear in many different counting problems. We can also obtain a closed form expression for the Catalan Sequence by using generating functions.

Theorem: If $a_0 = 1$, and $a_n = \sum_{k=1}^n a_{k-1} a_{n-k}$ for $n \geq 1$, then

$$\sum_{n \geq 0} a_n x^n = \frac{1 - \sqrt{1-4x}}{2x} \text{ and } a_n = \frac{1}{n+1} \binom{2n}{n}.$$

Key Definitions and Notation:

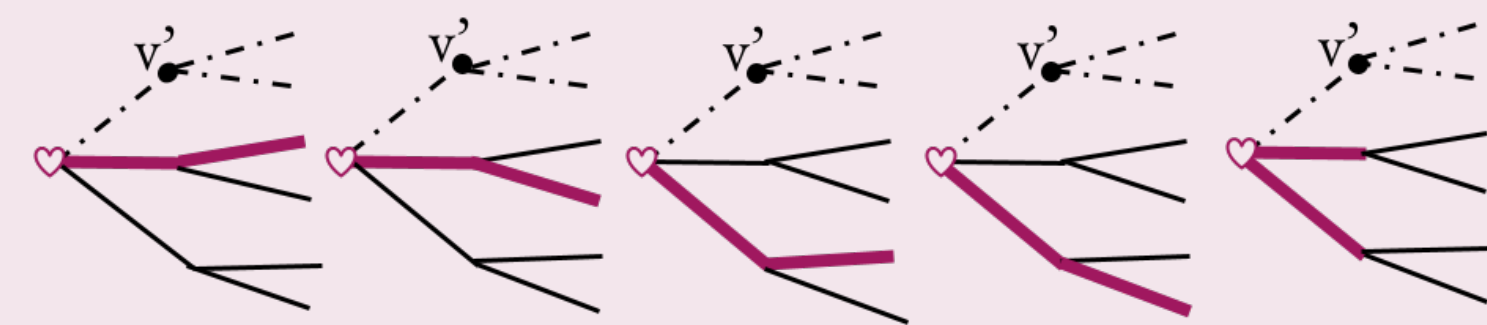
Having familiarized ourselves with generating functions, we will apply this knowledge to graph theory. We'll examine another recurrence relation, and use a generating functions to solve for an explicit formula. To setup our problem, we will introduce some terms:

- ***p*-percolation:** Given a graph G with vertex set $V(G)$, a ***p*-percolation** is a random graph on $V(G)$, where each edge in $E(G)$ has a probability p of being **open**, or included in the p -percolation, and a $1-p$ probability of being **closed**, or deleted.
- **Rooted Graph:** A rooted graph is a graph with a specified vertex $\mathbf{v}$ called the **root**.
- **Cluster size:** Given a ***p*-percolation** of a rooted graph G , the **cluster size**, denoted $K_p(G)$, is the random variable counting how many vertices remain in the same connected component of the root $\mathbf{v}$. This includes $\mathbf{v}$ itself.
- **Mass spectrum:** The mass spectrum is given by the probability distribution of $K_p(G)$.
- ***D*-regular tree:** A ***d*-regular tree** is a rooted infinite tree where every vertex is adjacent to d vertices. Up to isomorphism, there is only one d -regular tree, denoted V_d .

D-regular Trees (V_d): Setup

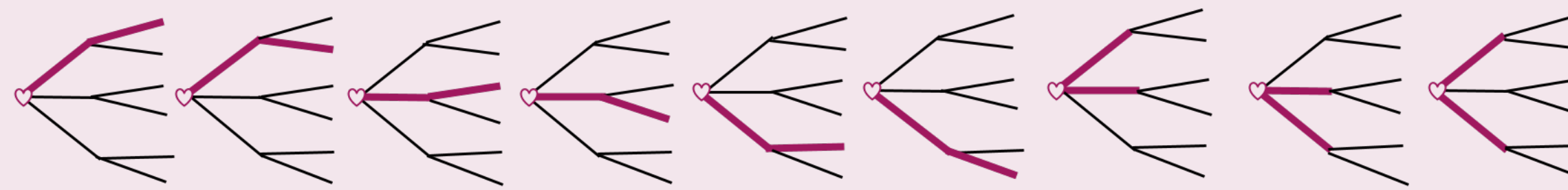
For the purpose of this poster, we will be considering d -regular trees, V_d . Our final goal will be to construct a mass spectrum for cluster size of p -percolation on V_d . We will do this by first constructing a recurrence relation for the number of ways a cluster size can occur in our p -percolation, and then use generating functions to obtain an explicit formula. We will conclude by using this formula to construct an equation for the probability of a cluster size $K_p(G)$. We first want to solve for the number of subtrees of a certain size connected to root v , so we define the following:

$C_{d,k}$: Fixing an arbitrary neighboring vertex v' of the root v , let $C_{d,k}$ be the number of subtrees of V_d with k vertices containing v but not containing v' . Note here that the value of $C_{d,k}$ is independent of our choice of v' due to the symmetry of V_d



Observe $C_{3,3} = 5$. Root vertex v denoted by ♥

$D_{d,k}$: Let this be the number of subtrees of V_d with k vertices containing v



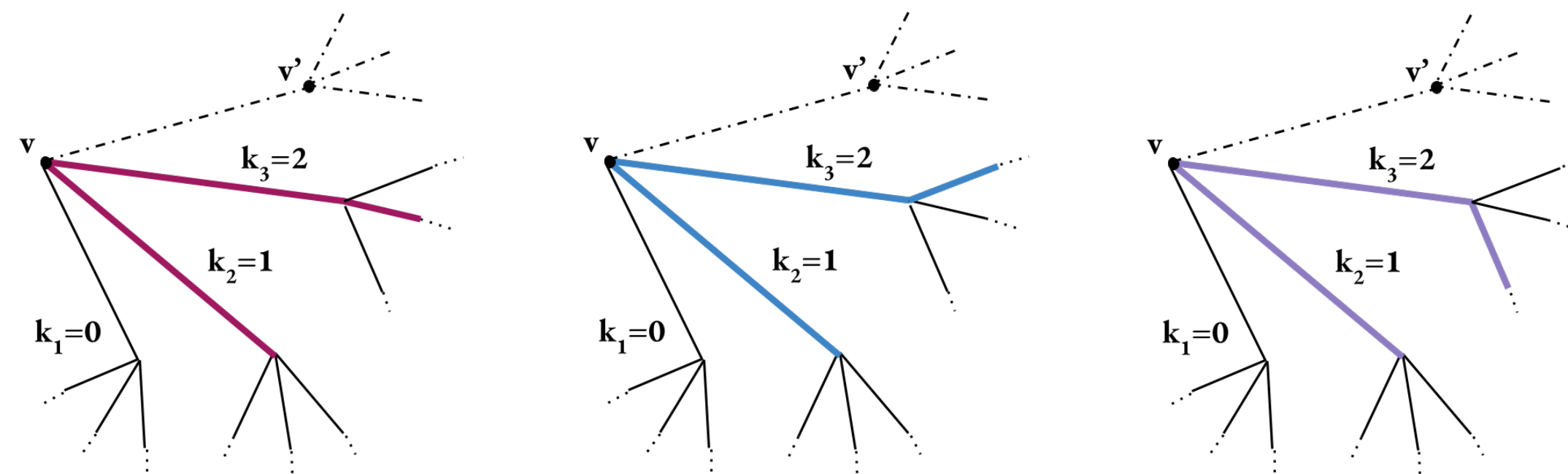
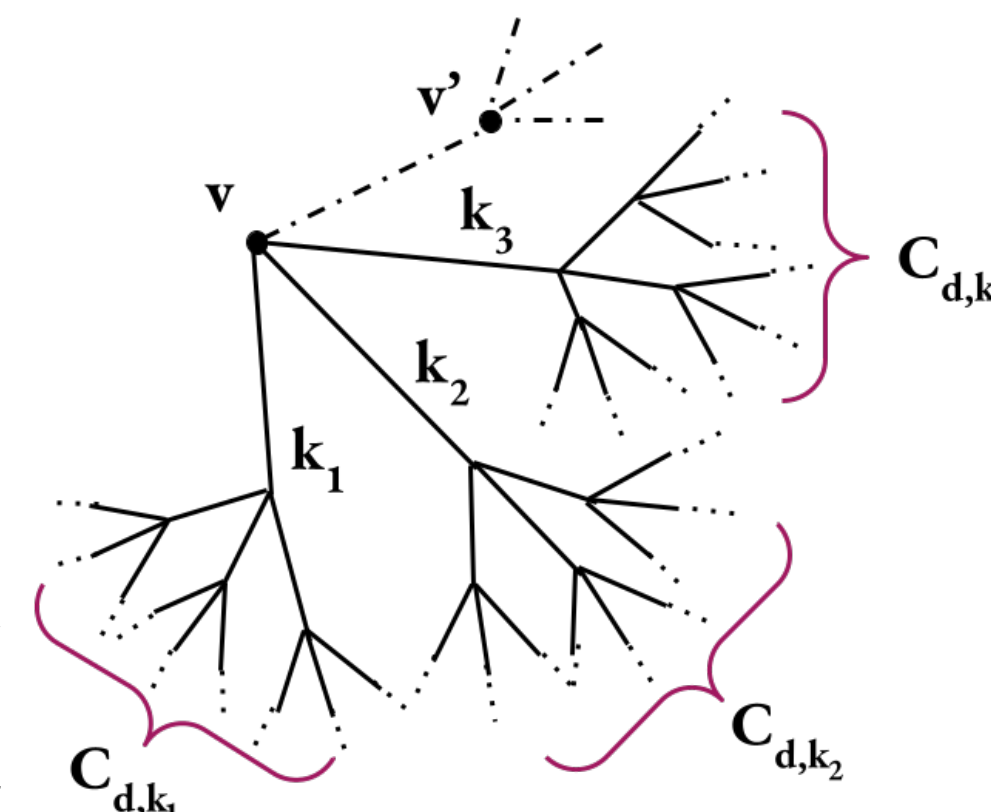
Observe that $D_{3,3} = 9$. Root vertex v denoted by ♥

D-regular Trees (V_d): Computing $C_{d,k}$

For example, let's start by considering V_4 and solving for $C_{4,4}$. We consider the set of all possible $(d-1)$ -tuples summing to $k-1=3$ given by:

$$K_{3,3} = \{(k_1, k_2, k_3) \in \mathbb{N}^3 : k_1 + k_2 + k_3 = k-1 = 3\}.$$

Essentially, the tuples tell us what size subtree we want from each immediate branch of the root, which we can visualize with the diagram to the right. Then we consider C_{d,k_i} for each branch and multiply $C_{d,k_1} * C_{d,k_2} * C_{d,k_3}$ to count all possible combinations of "good" subtrees for each $d-1$ -tuple. For example, for the tuple $(0, 1, 2)$, we want zero edges from the k_1 branch, one edge from the k_2 branch, and two edges from the k_3 branch. The three ways to construct a subtree as such are pictured below:



Summing over the tuples,

$$C_{4,4} = \sum_{(k_1, k_2, k_3) \in K_{3,3}} C_{d,k_1} * C_{d,k_2} * C_{d,k_3} = \sum_{(k_1, k_2, k_3) \in K_{3,3}} \prod_{i=1}^{d-1=3} C_{d,k_i}$$

D-regular Trees (V_d): Computing $C_{d,k}$

To generalize this example, to create a subtree of size k , we need $k-1$ edges. We consider elements, k' , of $K_{k-1,d-1} = \{(k_1, \dots, k_{d-1}) \in \mathbb{N}^{d-1} : \sum k_i = k-1\}$. Now for each branch, we use C_{d,k_i} to create subtrees of size k_i within each branch. Using the same reasoning we followed for V_4 , we can conclude:

Theorem: For $k \geq 1$, $C_{d,k}$ is recursively given by:

$$C_{d,k} = \sum_{k' \in K_{k-1,d-1}} \prod_{i=1}^{d-1} C_{d,k_i}$$

Now that we have obtained a recursive formula for $C_{d,k}$, we aim to use this to reach an explicit one. Let's setup generating function $C_d(t) = \sum_{k \geq 0} C_{d,k} t^k$ and observe:

$$C_d(t) = \sum_{k=0}^{\infty} C_{d,k} t^k = 1 + t \sum_{k=1}^{\infty} t^{k-1} \left(\sum_{(k_1, \dots, k_{d-1})} \prod_{i=1}^{d-1} C_{d,k_i} \right) = 1 + t C_d(t)^{d-1}$$

Lagrange Inversion Theorem

Suppose $Z_d(t)$ and $G(x)$ are power series, and $G(0) = 1$. If $tG(Z_d(t)) = Z_d(t)$, then

$$[t^k]Z_d(t) = \left(\frac{1}{k}\right) [Z_d(t)^{k-1}]G(Z_d(t))^k$$

Let $Z_d(t) = C_d(t) - 1$, notice $[t^k]Z_d(t) = [t^k]C_d(t)$ for $k \geq 1$. Now, let $G(x) = (x+1)^{d-1}$ so that $Z_d(t) = t(Z_d(t) + 1)^{d-1} = tG(Z_d(t))$. Now, we may apply Lagrange Inversion theorem to obtain that for $k \geq 1$,

$$C_{d,k} = [t^k]C_d(t) = [t^k]Z_d(t) = \left(\frac{1}{k}\right) [Z_d(t)^{k-1}]G(Z_d(t))^k = [x^{k-1}]G(x)^k = \binom{k(d-1)}{k-1}$$

Here, we can note that for 3-regular trees, $C_{3,k}$ gives the k^{th} Catalan number!

D-regular Trees (V_d): Computing $D_{d,k}$ and Mass Spectrum

Now that we've solved for an explicit formula for $C_{d,k}$, we can think of $D_{d,k}$ in terms of $C_{d,k}$ and simplify to obtain the following:

$$D_{d,k} = \sum_{(k_1, \dots, k_d)} \prod_{i=1}^d C_{d,k_i} = \sum_{s=0}^{k-1} \frac{\binom{s(d-1)}{s-1} \binom{(k-s)(d-1)}{k-s-1}}{s(k-s)}$$

Lastly, to construct a value for $\mathbb{P}(K_p(V_d) = k)$. Well, we can think of this as the sum of probabilities of occurrence for each tree counted by $D_{d,k}$. For a tree T counted by $D_{d,k}$, it has $k-1$ edges we want open in percolation, and $dk-2(k-1)$ edges connecting T to $V_d - T$ that we want closed. Multiplying this by $D_{d,k}$ since each tree counted by $D_{d,k}$ has equal chance of occurrence, we conclude:

$$\mathbb{P}(K_p(V_d) = k) = (D_{d,k}) p^{k-1} (1-p)^{dk-2(k-1)}$$

References

Bonnano, Leo, Sawyer Dobson, Juyon Lee, and Titus Sharman. Percolation on Graphs. Apr. 2024.

West, Douglas Brent. Combinatorial Mathematics. Cambridge University Press, 2021.

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